



Title

rdms — Point Estimation and Robust Bias-Corrected Inference for Multi-Score Designs.

Syntax

```
rdms depvar runvar1 [runvar2 treatvar] [if] [in], cvar(cvar1 [cvar2]) [
  range(range1 [range2]) xnorm(string) fuzzy(string) derivvar(string)
  pooled_opt(string) pvar(string) qvar(string) hvar(string) hrightvar(string)
  bvar(string) brightvar(string) rho(string) covs(string)
  covs_dropvar(string) kernelvar(string) weightsvar(string) bwselectvar(string)
  scaleparvar(string) scaleregulvar(string) masspointsvar(string)
  bwcheckvar(string) bwrestrictvar(string) stdvarsvar(string) vcevar(string)
  level(#) all detail vleverage nochecks nowarnings plot graph_opt(string)
  conventional ]
```

Description

rdms implements point estimation and robust bias-corrected inference for regression discontinuity (RD) designs with multiple scores. For methodological background see [Keele and Titiunik \(2015\)](#), [Cattaneo, Keele, Titiunik and Vazquez-Bare \(2016\)](#), and [Cattaneo, Keele, Titiunik and Vazquez-Bare \(2021\)](#). It also computes alternative estimation and inference procedures available in the literature.

If only *runvar1* is specified, **rdms** analyzes an RD design with cumulative cutoffs in which a unit gets different dosages of a treatment depending on the value of *runvar1*. If *runvar1*, *runvar2* and *treatvar* are specified, **rdms** analyzes an RD design with two running variables in which units with *treatvar* equal to one are treated.

Companion commands are: **rdmc** for point estimation and robust bias-corrected inference for multi-cutoff designs, and **rdmcpplot** for data-driven RD plots for multi-cutoff designs.

A detailed introduction to this command is given in [Cattaneo, Titiunik and Vazquez-Bare \(2020\)](#).

Companion R functions are also available [here](#).

This command employs the Stata (and R) package **rdrobust** for underlying calculations. See [Calonico, Cattaneo and Titiunik \(2014\)](#), [Calonico, Cattaneo and Titiunik \(2015\)](#), and [Calonico, Cattaneo, Farrell and Titiunik \(2017\)](#) for more details.

Related Stata and R packages useful for inference in RD designs are described in the following website:

<https://rdpackages.github.io/>

Options

Estimand

cvar(string) specifies the numeric variable *cvar1* containing the RD cutoff for *indepvar* in a cumulative cutoffs setting, or the two scores *cvar1* and *cvar2* in a two-score setting.

range(range1 [range2]) specifies the range of the running variable to be used for estimation around each cutoff. Specifying only one variable implies using the same range at each side of the cutoff.

xnorm(string) specifies the normalized running variable to estimate pooled effect.

fuzzy(string) indicates a fuzzy design. Use **fuzzy**(*treatvar* *sharpbw*) to request sharp-RD bandwidth selection for fuzzy RD. See [rdrobust](#) for details.

derivvar(*string*) a variable of length equal to the number of different cutoffs that specifies the order of the derivative for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

rdrobust Options

pooled_opt(*string*) specifies the options to be passed to **rdrobust** to calculate pooled estimates. See [rdrobust](#) for details.

Local Polynomial Regression

pvar(*string*) a variable of length equal to the number of different cutoffs that specifies the order of the polynomials for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

qvar(*string*) a variable of length equal to the number of different cutoffs that specifies the order of the polynomials for bias estimation for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

hvar(*string*) a variable of length equal to the number of different cutoffs that specifies the bandwidths for **rdrobust** to calculate cutoff-specific estimates. When **hrightvar** is specified, **hvar** indicates the bandwidth to the left of the cutoff. When **hrightvar** is not specified, the same bandwidths are used at each side. See [rdrobust](#) for details.

hrightvar(*string*) a variable of length equal to the number of different cutoffs that specifies the bandwidths to the right of the cutoff for **rdrobust** to calculate cutoff-specific estimates. When **hrightvar** is not specified, the bandwidths in **hvar** are used at each side. See [rdrobust](#) for details.

bvar(*string*) a variable of length equal to the number of different cutoffs that specifies the bandwidths for bias estimation for **rdrobust** to calculate cutoff-specific estimates. When **brightvar** is specified, **bvar** indicates the bandwidth to the left of the cutoff. When **brightvar** is not specified, the same bandwidths are used at each side. See [rdrobust](#) for details.

brightvar(*string*) a variable of length equal to the number of different cutoffs that specifies the bandwidths for bias estimation to the right of the cutoff for **rdrobust** to calculate cutoff-specific estimates. When **brightvar** is not specified, the bandwidths in **bvar** are used at each side. See [rdrobust](#) for details.

rho**var**(*string*) a variable of length equal to the number of different cutoffs that specifies the value of rho for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

covs**var**(*string*) a variable of length equal to the number of different cutoffs that specifies the covariates for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

covs_drop**var**(*string*) a variable of length equal to the number of different cutoffs that specifies whether collinear covariates should be dropped. See [rdrobust](#) for details.

kernel**var**(*string*) a variable of length equal to the number of different cutoffs that specifies the kernels for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

weights**var**(*string*) a variable of length equal to the number of different cutoffs that specifies the weights for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

Bandwidth Selection

bwselect**var**(*string*) a variable of length equal to the number of different cutoffs that specifies the bandwidth selection method for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

scaleparvar(string) a variable of length equal to the number of different cutoffs that specifies the value of scalepar for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

scaleregulvar(string) a variable of length equal to the number of different cutoffs that specifies the value of scaleregul for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

masspointsvar(string) a variable of length equal to the number of different cutoffs that specifies how to handle repeated values in the running variable. See [rdrobust](#) for details.

bwcheckvar(string) a variable of length equal to the number of different cutoffs that specifies the value of **bwcheck**. See [rdrobust](#) for details.

bwrestrictvar(string) a variable of length equal to the number of different cutoffs that specifies whether computed bandwidths are restricted to the range of **runvar**. See [rdrobust](#) for details.

stdvarsvar(string) a variable of length equal to the number of different cutoffs that specifies whether **depvar** and **runvar** are standardized. See [rdrobust](#) for details.

Variance-Covariance Estimation and Inference

vcevar(string) a variable of length equal to the number of different cutoffs that specifies the variance-covariance matrix estimation method for **rdrobust** to calculate cutoff-specific estimates. Current options include **nn**, **hc0**, **hc1**, **hc2**, **hc3**, **cr1**, **cr2**, and **cr3**; cluster-robust options include the cluster variable in the same string, for example **cr2 schoolid**. See [rdrobust](#) for details.

level(#) specifies the confidence level for confidence intervals. See [rdrobust](#) for details.

all, **detail**, and **vleverage** pass the corresponding display and diagnostic options to **rdrobust**.

nochecks and **nowarnings** pass the corresponding validation and warning controls to **rdrobust**.

conventional displays conventional, instead of robust-bias corrected, p-values and confidence intervals.

Plot

plot plots the pooled and cutoff-specific estimates.

graph_opt(string) options to be passed to the graph when **plot** is specified.

Examples

```
Standard use of rdms for cumulative cutoffs
. rdms yvar xvar, c(cvar)
```

```
rdms with plot
. rdms yvar xvar, c(cvar) plot
```

```
Standard use of rdms for multiple scores
. rdms yvar xvar1 xvar2 treatvar, c(cvar)
```

Saved results

rdms saves the following in **e()**:

Matrices

e(b)	bias-corrected coefficient vector
e(V)	robust variance-covariance matrix of the estimators
e(coefs)	conventional coefficient vector
e(pv_rb)	robust p-value vector
e(CI_rb)	bias-corrected confidence intervals
e(H)	vector of bandwidths at each side of each cutoff
e(sampsis)	vector of sample sizes at each side of each cutoff

References

- Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik. 2017. rdrobust: Software for Regression Discontinuity Designs. *Stata Journal* 17(2): 372-404.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014. Robust Data-Driven Inference in the Regression-Discontinuity Design. *Stata Journal* 14(4): 909-946.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2015. rdrobust: An R Package for Robust Nonparametric Inference in Regression-Discontinuity Designs. *R Journal* 7(1): 38-51.
- Cattaneo, M. D., L. Keele, R. Titiunik, and G. Vazquez-Bare. 2016. Interpreting Regression Discontinuity Designs with Multiple Cutoffs. *Journal of Politics* 78(4): 1229-1248.
- Cattaneo, M. D., L. Keele, R. Titiunik, and G. Vazquez-Bare. 2021. Extrapolating Treatment Effects in Multi-Cutoff Regression Discontinuity Designs. *Journal of American Statistical Association* 116(536): 1941-1952.
- Cattaneo, M. D., R. Titiunik, and G. Vazquez-Bare. 2020. Analysis of Regression Discontinuity Designs with Multiple Cutoffs or Multiple Scores. *Stata Journal* 20(4): 866-891.
- Keele, L., and R. Titiunik. 2015. Geographic Boundaries as Regression Discontinuities. *Political Analysis* 23(1): 127-155.

Authors

- Matias D. Cattaneo, Princeton University, Princeton, NJ.
matias.d.cattaneo@gmail.com.
- Rocio Titiunik, Princeton University, Princeton, NJ. rocio.titiunik@gmail.com.
- Gonzalo Vazquez-Bare, UC Santa Barbara, Santa Barbara, CA.
gvazquezbare@gmail.com.